Youssef Yasser Elsayed Hamed

AI & Software Engineer

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Profile

AI & Software Engineer with approximately six months of internship and freelance experience, building computer vision and generative AI applications using Python, PyTorch, and FastAPI. Work includes real-time object detection and tracking, deep learning model training, and RAG applications. Combines ML and backend skills to build and integrate AI-powered services.

Education

Cairo University Cairo, Egypt
Bachelor's in Computer Science and Artificial Intelligence *Sep. 2023 – July 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Introduction to Software Engineering, Signals & Systems, Generative Adversarial Networks (GANs), Supervised & Unsupervised Learning, Reinforcement Learning, Robotics & Autonomous Multiagent Systems, Probability & Statistics

Professional Experience

AI Training Evaluator (Freelance) Jun. 2026 – Aug. 2026
Outlier *Remote, Cairo, Egypt*

- Evaluated **500+** conversational AI outputs for **RLHF** pipelines, maintaining a **95%+** QA score across naturalness and utility.

Aitronix Internship On-site, New Cairo, Egypt
Artificial Intelligence Intern | *Certificate* *Jun. 2026 – Aug. 2026*

- Developed production-level computer vision and LLM systems, leveraging **FastAPI** to optimize backend data processing times by **30%**.
- Deployed an **n8n**-powered **WhatsApp** bot processing **150+** daily inquiries, reducing manual response latency by **80%**.

Computer Vision Intern *Apr. 2026 – Jun. 2026*

- Built and optimized real-world CV pipelines, integrating **Vision Science** and **RAG** for enhanced context retrieval.
- Engineered a **Behavioral AI Theft Detection** system, classifying suspicious interactions with **90%** accuracy.
- Deployed **YOLOv8s** models onto enterprise IP cameras via RTSP, achieving **25+ FPS** spatial monitoring across **4** concurrent streams.

Apprenticeships & Technical Training

AI Trainee Huawei ICT Academy Program | *Certificate* Sep. 2026 – Oct. 2026
Huawei ICT Academy-Egypt *Remote, Cairo, Egypt*

- Completed the intensive Huawei HCIA-AI training program, covering **5+** core modules across the full AI development stack.

Computer Vision Engineer Trainee | *Certificate* Jul. 2026 – Aug. 2026
National Telecommunication Institute (NTI) *Hybrid, Cairo, Egypt*

- Collaborated on assembling a dataset of **10,000+ images** from multiple datasets available on **Roboflow** for driver fatigue and attention monitoring.
- Developed a **Driver Fatigue & Attention Monitoring System [GitHub]**, detecting drowsiness in real time at **30+ FPS** using **MediaPipe** and **MobileNetV2**.
- Built a **Flask/SQLite** dashboard and implemented temporal tracking, reducing false alarms caused by normal blinking by **40%**.

Technical Skills

Programming & Fundamentals: Python, C++, SQL, Java, JavaScript, C#; OOP, Design Patterns, Git (Version Control)

Machine Learning & Deep Learning: PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face, Pandas, Numpy

Computer Vision: OpenCV, YOLO, Multi-Object Tracking (ByteTrack), Pose Estimation (MediaPipe), Image Segmentation, Small-Object Detection (SAHI), Anomaly Detection

Data Processing: Data Cleaning, Data Preprocessing, Feature Engineering, Data Augmentation

Generative AI: LLMs, NLP, Vision-Language Models, RAG, Agentic Workflows, LangChain, LlamaIndex, Vector Databases (ChromaDB, Pinecone)

Backend & Databases: FastAPI, REST APIs, WebSockets, PostgreSQL, Supabase

Deployment & MLOps: Docker, Kubernetes, GitHub Actions, MLflow, ONNX, TensorRT; AWS SageMaker, Azure

Projects

Pie: AI-Powered Marketing Kit Generator | **[Live Demo]** | *VLMs, Groq LPU, RAG, FastAPI* August 2026

- **Generative AI & RAG:** Built a full-stack AI platform using **Vision-Language Models (VLMs)** to transform product screenshots into structured marketing kits in under **60 seconds**.
- **High-Speed Inference:** Integrated **Groq-hosted LLM inference** with a custom **RAG** pipeline to retrieve contextual information for marketing content generation.

Botfolio: AI Portfolio Chatbot Platform | **[Live Demo]** | *Python, FastAPI, FAISS, Groq, SQLite* June 2026

- Built a multi-tenant AI chatbot platform used by 10 users, with JWT authentication, password hashing, resume uploads, customizable bots, and shareable public chat links.
- Implemented RAG with per-chatbot FAISS indexes and Groq-powered responses, SQLite data storage, Stripe Checkout and billing webhooks, and an HTML/CSS/JavaScript dashboard for chatbot configuration and user/plan management.

YAI : Video Safety Analyzer | **[GitHub] [Live Demo]** | *PyTorch, FastAPI, YOLOv10x, CLIP* May 2026

- Built a five-stage video analysis pipeline combining object detection, open-vocabulary classification, multi-object tracking, behavioral anomaly scoring across six threat categories, and monocular depth estimation.
- Integrated YOLOv10x, CLIP, ByteTrack, AnomalyTransformer, and Depth Anything v2 through an asynchronous FastAPI backend with job queues and WebSocket result streaming to a React frontend hosted on Netlify.

Smart Agriculture Crop Disease Detection | **[Kaggle]** | *ViT, Attention-CNN, DCGAN, PyTorch* Feb - May 2026

- **Automated Diagnosis & Kaggle Bronze Medal:** Built a robust disease classification system trained on **54,000+** images across **15** categories. Benchmarked **14** architectures, with a custom **CNN** achieving **97.5%** validation accuracy. Awarded a **Kaggle Bronze Medal** for the comprehensive model evaluation notebook.

Behavioral AI Theft Detection System | **[GitHub]** | *YOLOv11x, SAHI, ResNet18 ReID, LSTM* April 2026

- **Robust Detection & Tracking:** Engineered a real-time pipeline actively monitoring **30** high-value classes using **YOLOv11x** and **SAHI** (**1000**×**1000** tiling), integrated with **ByteTrack** and **ResNet18** ReID.
- **Behavioral Classification:** Extracted **17** dynamic spatial features (grab motion, joint angles) to train an attention-based PyTorch **LSTM** network, classifying suspicious temporal movements across occlusions.

Certificates & Courses

Huawei Certified ICT Associate (HCIA-AI V4.0) | *Huawei ICT Academy (ID: ICT20261004014238)* Oct. 2026
Microsoft Applied Skills: Secure AI Solutions, Cloud Security & Entra ID | *Microsoft* Sep. 2026
Google AI Essentials | *Google (ID: 8C77QG2AJQAE)* Sep. 2026
Claude Code in Action | *Anthropic (ID: 3i22h3mb75cs)* June 2026
Getting Started with Deep Learning | *NVIDIA (ID: e4Sr2s2OTXC-nnKTTv2bqQ)* Dec. 2025
Master OpenCV Fundamentals for Real-Time CV | *Coursera – EDUCBA* May 2026
Deep Learning Specialization | *DeepLearning.AI (Andrew Ng) – Coursera* Feb. 2026

Competitions

Competitive Programmer | *Certificate* July 2024
Egyptian Collegiate Programming Contest (ECPC) *Alexandria, Egypt*

- Engineered optimized algorithmic solutions in **C++** under strict time constraints, applying advanced data structures to tackle complex computational problems.
- Collaborated in a high-pressure team environment to prioritize problem selection, rapidly debug logical errors, and maximize competitive performance.

Languages

Arabic: Native
English: Professional Working Proficiency (B2+ / Duolingo English Test: 130)